

Artificial Intelligence (CS-203)

Objectives:

- **Proposition**
 - **Compound Proposition**
 - **Logical Equivalence**
 - **Inference Rules**
-

Proposition

A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Example 1: All of the following declarative sentences are propositions.

- | | |
|--|--------------|
| 1) Washington, D.C., is the capital of the United States of America. | True |
| 2) Toronto is the capital of Canada. | False |
| 3) $1 + 1 = 2$. | True |
| 4) $2 + 2 = 3$. | False |

Example 2: Consider the following sentences. These are not propositions

- 1) What time is it?
- 2) Read this carefully.
- 3) $x + 1 = 2$.
- 4) $x + y = z$.

Propositional Variables

Letters are used to denote **propositional variables** or **statement variables**. Conventional letters used for propositional variables are $p, q, r, s, \dots$. The **truth value** of a proposition is denoted by **T**, if it is a true proposition, and the truth value of a proposition is denoted by **F**, if it is false proposition.

Propositional Logic

The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**.

Compound proposition

New propositions, called **compound propositions**, are formed from existing propositions using logical operators. The propositional operators according to precedence are given below.

Negation $\neg$

Conjunction $\wedge$

Disjunction $\vee$

Implication $\rightarrow$

Biconditional $\leftrightarrow$

1) Negation operator

It is a unary operator which reverses the truth value of its operand. The proposition $\neg p$ is read as “not p .” The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .

Example

P: It is hot today.

$\neg P$: It is not hot today.

Truth table

P	$\neg P$
T	F
F	T

2) Conjunction

It is a binary operator which produces a compound proposition which is true if and only if both of its operands are true, otherwise it is false. Let p and q be propositions. The *conjunction* of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true.

Example

P: Age is more than 20 years.

Q: Student passed intermediate exam.

$P \wedge Q$: Age is greater than 20 years and student has passed intermediate.

Truth table

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

3) Disjunction

It is a binary operator which produces a compound proposition which is false if and only if both of the operands are false otherwise it is true. Let p and q be propositions. The *disjunction* of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Example

P: Student has passed calculus.

Q: Student has passed the programming.

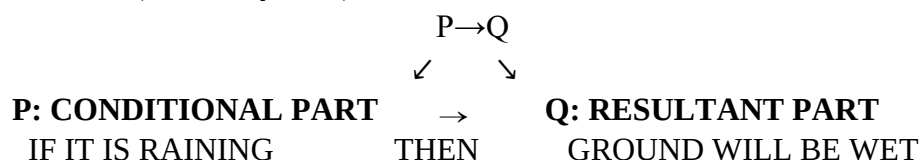
$P \vee Q$: Student has passed programming or calculus.

Truth table

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

4) Implication operator/ Conditional Statements

Let p and q be propositions. The *conditional statement* $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the *hypothesis* (or *antecedent* or *premise*) and q is called the *conclusion* (or *consequence*).



Truth table

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

CONVERSE, CONTRAPOSITIVE, AND INVERSE

We can form some new conditional statements starting with a conditional statement $p \rightarrow q$. In particular, there are three related conditional statements that occur so often that they have special names.

The proposition $q \rightarrow p$ is called the *converse* of $p \rightarrow q$.

The proposition $\neg q \rightarrow \neg p$ is the *contrapositive* of $p \rightarrow q$.

The proposition $\neg p \rightarrow \neg q$ is called the *inverse* of $p \rightarrow q$.

5) Biconditional $\leftrightarrow$

Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$ is the proposition “ p if and only if q .” The statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called *bi-implications*.

Example

P: Take the flight.

Q: Buy a ticket.

$P \leftrightarrow Q$ “You can take the flight if and only if you buy a ticket.”

Truth table

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

A comparison table of all logical operators is given below

Operator/ connective symbols	Words	Technical term	Precedence
$\neg$	NOT	Negation	1
$\wedge$	AND	Conjunction	2
$\vee$	OR	Disjunction	3
$\rightarrow$	Implies	Implication	4
$\leftrightarrow$	If and only if	biconditional	5

Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- It is below freezing and snowing.
- It is below freezing but not snowing.
- It is not below freezing and it is not snowing.
- It is either snowing or below freezing (or both).
- If it is below freezing, it is also snowing.
- Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.

Logical Equivalences

Compound propositions that have the same truth values in all possible cases are called **logically equivalent**. One way to determine whether two compound propositions are equivalent is to use a truth table. The compound propositions p and q are equivalent if and only if the columns giving their truth values are same. Any of the following logically equivalent expressions may be used to replace each other, wherever they occur:

Equivalence	Names
$p \vee \neg p \equiv T$	Tautology
$p \wedge \neg p \equiv F$	Contradiction
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg (p \wedge q) \equiv \neg p \vee \neg q$ $\neg (p \vee q) \equiv \neg p \wedge \neg q$	De Morgans laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorbtion laws
$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation laws
$p \rightarrow q \equiv \neg p \vee q$	Implication
$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$	Equivalence
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	Transportation

Inference Rules

In artificial intelligence, we need intelligent computers which can create new logic from old logic or by evidence, **so generating the conclusions from evidence and facts is termed as Inference.** Inference rules can be applied to logical arguments to prove their validity or invalidity, i.e. carry out deduction.

Rules of Inference	Tautological Form	Name
P ----- $P \vee Q$	$P \Rightarrow (P \vee Q)$	addition
$P \wedge Q$ ----- P	$(P \wedge Q) \Rightarrow P$	simplification
P $P \rightarrow Q$ ----- Q	$[P \wedge (P \rightarrow Q)] \Rightarrow Q$	modus ponens
$\neg Q$ $P \rightarrow Q$ ----- $\neg P$	$[\neg Q \wedge (P \rightarrow Q)] \Rightarrow \neg P$	modus tollens

$\begin{array}{l} P \vee Q \\ \neg P \\ \hline Q \end{array}$	$[(P \vee Q) \wedge \neg P] \Rightarrow Q$	disjunctive syllogism
$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \\ \hline P \rightarrow R \end{array}$	$[(P \rightarrow Q) \wedge (Q \rightarrow R)] \Rightarrow [P \rightarrow R]$	hypothetical syllogism
$\begin{array}{l} P \\ Q \\ \hline P \wedge Q \end{array}$		conjunction
$\begin{array}{l} (P \rightarrow Q) \wedge (R \rightarrow S) \\ P \vee R \\ \hline Q \vee S \end{array}$	$[(P \rightarrow Q) \wedge (R \rightarrow S) \wedge (P \vee R)] \Rightarrow [Q \vee S]$	constructive dilemma
$\begin{array}{l} (P \rightarrow Q) \wedge (R \rightarrow S) \\ \neg Q \vee \neg S \\ \hline \neg P \vee \neg R \end{array}$	$[(P \rightarrow Q) \wedge (R \rightarrow S) \wedge (\neg Q \vee \neg S)] \Rightarrow [\neg P \vee \neg R]$	destructive dilemma

Formal Proof of Validity:**Example1:**

The formal strategy to carry out a proof is as follows: Lets assume that we are given the following argument, and we have to prove its validity.

$$\begin{array}{l} A \Rightarrow B \\ B \Rightarrow C \\ C \Rightarrow D \\ \neg D \\ A \vee E \\ \therefore E \end{array}$$

First we list all the premises and assign them numbers on the left as follows:

$$\begin{array}{ll} (1) & A \Rightarrow B \\ (2) & B \Rightarrow C \\ (3) & C \Rightarrow D \\ (4) & \neg D \\ (5) & A \vee E \end{array} \quad \therefore E$$

Notice that in line 5 we have also given the conclusion, which we want to derive. After giving the premises we carry out the proof by applying inference rules and replacement rules, in the subsequent lines and give them numbers on the left. On the right of these lines we also give the line numbers from which this new line was arrived at by applying a rule:

- | | | |
|-----|-------------------|-------------------------------|
| (1) | $A \Rightarrow B$ | |
| (2) | $B \Rightarrow C$ | |
| (3) | $C \Rightarrow D$ | |
| (4) | $\neg D$ | |
| (5) | $A \vee E$ | $\therefore E$ |
| (6) | $A \Rightarrow C$ | 1, 2, Hypothetical syllogism. |
| (7) | $A \Rightarrow D$ | 6, 3, Hypothetical syllogism. |
| (8) | $\neg A$ | 7, 4, Modus Tollens. |
| (9) | E | 5, 8, Disjunctive Syllogism. |

Line 6 was arrived at by using premises 1, 2 and applying Hypothetical syllogism rule. Line 7 was also arrived at by applying Hypothetical syllogism on lines 6 and 3. For line 8 we used Modus Tollens on lines 7 and 4, which was then used in line 9 to prove the conclusion by applying Disjunctive syllogism.

Example 2:

1. Today is Tuesday or Wednesday.
2. But it can't be Wednesday, since the doctor's office is open today, and that office is always closed on Wednesdays.
3. Therefore today must be Tuesday.

First of all we will define propositional variables for them

P= Today is Tuesday.

Q= Today is Wednesday

R= The doctor's office is open today

S= The doctor's office is always closed on Wednesdays

Rewrite arguments using logical operators

- (1) $P \vee Q$
- (2) $R \wedge S \Rightarrow \neg Q$, (where we can replace S with $Q \Rightarrow \neg R$)
- (3) $\therefore P$

Now premises becomes

- (1) $P \vee Q$

(2) $R \wedge (Q \Rightarrow \neg R) \Rightarrow \neg Q$ This is Correct by Modus Tollens

(3) $\therefore P$

(4) P 1, 2 Disjunctive syllogism
